ADVANCED PYTHON PROGRAMMING

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Summaries – Practice

Prompt 1:
“Rank
City
Green View Index (%)

1	Tampa	36.1
2	Breda	29.3
3	Singapore	29.3
4	Oslo	28.8
5	Sydney	25.9
6	Vancouver	25.9
7	Montreal	25.5
8	Cambridge	25.3
9	Xalapa	24.0
10	Durban	23.7
11	Johannesburg	23.6
12	Sacramento	23.6
13	Frankfurt	21.5
14	Geneva	21.4
15	Veracruz	21.4
16	Amsterdam	20.6
17	Seattle	20.0

Rank
City
Green View Index (%)

18	Toronto	19.5
19	Miami	19.4
20	Boston	18.2
21	Tel Aviv	17.5
22	Guadalajara	17.3
23	León	16.5
24	Turin	16.2
25	Los Angeles	15.2
26	Buenos Aires	14.5
27	Franca	13.7
28	New York	13.5
29	Cape Town	13.4
30	London	12.7
31	São Paulo	11.7
32	Quito	10.8
33	Kobe	9.4
34	Paris	8.8

i want the dictionary consists of ranks, cities, its countries and its gvi index from the given table in python”

Prompt 2: “I created a dataframe "df" of gvi_data.
Now, Give me the python code for finding the average of GVI of USA from df”

Prompt 3: “Give me python code to find Sum of GVI of Cities of "Italy"”

Prompt 4: “How to find the country that has maximum number of cities in the dataframe”

Prompt 5: “How to find all these using pivot table”

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SUMMARIES PRACTICE

In [8]:
```
import pandas as pd

gvi_data = {
    "Rank": list(range(1, 35)),
    "City": [
        "Tampa", "Breda", "Singapore", "Oslo", "Sydney", "Vancouver", "Mo
        "Johannesburg", "Sacramento", "Frankfurt", "Geneva", "Veracruz",
        "Tel Aviv", "Guadalajara", "León", "Turin", "Los Angeles", "Bueno
        "São Paulo", "Quito", "Kobe", "Paris"
    ],
    "Country": [
        "USA", "Netherlands", "Singapore", "Norway", "Australia", "Canada
        "South Africa", "USA", "Germany", "Switzerland", "Mexico", "Nethe
        "Israel", "Mexico", "Mexico", "Italy", "USA", "Argentina", "Brazi
        "Brazil", "Ecuador", "Japan", "France"
    ],
    "GVI_Index_%": [
        36.1, 29.3, 29.3, 28.8, 25.9, 25.9, 25.5, 25.3, 24.0, 23.7,
        23.6, 23.6, 21.5, 21.4, 21.4, 20.6, 20.0, 19.5, 19.4, 18.2,
        17.5, 17.3, 16.5, 16.2, 15.2, 14.5, 13.7, 13.5, 13.4, 12.7,
        11.7, 10.8, 9.4, 8.8
    ]
}
```

In [9]:
```
df = pd.DataFrame(gvi_data)
df
```

Out[9]:

	Rank	City	Country	GVI_Index_%
0	1	Tampa	USA	36.1
1	2	Breda	Netherlands	29.3
2	3	Singapore	Singapore	29.3
3	4	Oslo	Norway	28.8
4	5	Sydney	Australia	25.9
5	6	Vancouver	Canada	25.9
6	7	Montreal	Canada	25.5
7	8	Cambridge	USA	25.3
8	9	Xalapa	Mexico	24.0
9	10	Durban	South Africa	23.7
10	11	Johannesburg	South Africa	23.6
11	12	Sacramento	USA	23.6
12	13	Frankfurt	Germany	21.5
13	14	Geneva	Switzerland	21.4
14	15	Veracruz	Mexico	21.4
15	16	Amsterdam	Netherlands	20.6
16	17	Seattle	USA	20.0
17	18	Toronto	Canada	19.5
18	19	Miami	USA	19.4
19	20	Boston	USA	18.2
20	21	Tel Aviv	Israel	17.5
21	22	Guadalajara	Mexico	17.3
22	23	León	Mexico	16.5
23	24	Turin	Italy	16.2
24	25	Los Angeles	USA	15.2
25	26	Buenos Aires	Argentina	14.5
26	27	Franca	Brazil	13.7
27	28	New York	USA	13.5
28	29	Cape Town	South Africa	13.4
29	30	London	UK	12.7
30	31	São Paulo	Brazil	11.7
31	32	Quito	Ecuador	10.8
32	33	Kobe	Japan	9.4
33	34	Paris	France	8.8

Average of GVI of "USA"

In [10]:
```
# Filter the DataFrame for 'USA' and calculate the average GVI
avg_gvi_usa = df[df['Country'] == 'USA']['GVI_Index_%'].mean()

print(f"The average GVI of the USA is: {avg_gvi_usa:.2f}%")
```

```
The average GVI of the USA is: 21.41%
```

Minimum of GVI of "Netherlands"

In [11]:
```
min_gvi_netherlands = df[df['Country'] == 'Netherlands']['GVI_Index_%'].m

print(f"The minimum GVI of the country Netherlands : {min_gvi_netherlands
```

```
The minimum GVI of the country Netherlands : 20.6%
```

Sum of GVI of Cities of "Italy"

In [12]:
```
sum_gvi_italy = df[df['Country'] == 'Italy']['GVI_Index_%'].sum()

print(f"The sum GVI of the country Italy : {sum_gvi_italy}%")
```

```
The sum GVI of the country Italy : 16.2%
```

Country that has maximum number of cities in count

In [13]:
```
# Finds the name of the country with the most cities
top_country = df['Country'].value_counts().idxmax()

print(f"The country with the maximum number of cities is: {top_country}")
```

```
The country with the maximum number of cities is: USA
```

Using Pivot Table

In [15]:
```
# 1. Create a pivot table summarizing every country across all core metri
gvi_pivot = df.pivot_table(
    index='Country',
    values=['GVI_Index_%', 'City'],
    aggfunc={'GVI_Index_%': ['mean', 'sum', 'min'], 'City': 'count'}
)

# Clean up column names for readability
gvi_pivot.columns = ['Total_Cities', 'Min_GVI', 'Average_GVI', 'Sum_GVI']

# 2. Display the entire summary table
print("--- Country Summary Pivot Table ---")
print(gvi_pivot)
```

```
--- Country Summary Pivot Table ---
              Total_Cities    Min_GVI  Average_GVI  Sum_GVI
Country
Argentina                1  14.500000         14.5     14.5
Australia                1  25.900000         25.9     25.9
Brazil                   2  12.700000         11.7     25.4
Canada                   3  23.633333         19.5     70.9
Ecuador                  1  10.800000         10.8     10.8
France                   1   8.800000          8.8      8.8
Germany                  1  21.500000         21.5     21.5
Israel                   1  17.500000         17.5     17.5
Italy                    1  16.200000         16.2     16.2
Japan                    1   9.400000          9.4      9.4
Mexico                   4  19.800000         16.5     79.2
Netherlands              2  24.950000         20.6     49.9
Norway                   1  28.800000         28.8     28.8
Singapore                1  29.300000         29.3     29.3
South Africa             3  20.233333         13.4     60.7
Switzerland              1  21.400000         21.4     21.4
UK                       1  12.700000         12.7     12.7
USA                      8  21.412500         13.5    171.3
```

In [16]:
```python
# Extract all 4 answers from the pivot table 'gvi_pivot'
avg_usa = gvi_pivot.loc['USA', 'Average_GVI']
min_netherlands = gvi_pivot.loc['Netherlands', 'Min_GVI']
sum_italy = gvi_pivot.loc['Italy', 'Sum_GVI']
top_country = gvi_pivot['Total_Cities'].idxmax()

# Print out the results cleanly
print(f"1. Average GVI of USA: {avg_usa:.2f}%")
print(f"2. Minimum GVI of the country in the 2nd Rank (Netherlands): {min
print(f"3. Sum of GVI of Cities of Italy: {sum_italy:.1f}%")
print(f"4. Country with the maximum number of cities: {top_country}")
```

```
1. Average GVI of USA: 13.50%
2. Minimum GVI of the country in the 2nd Rank (Netherlands): 25.0%
3. Sum of GVI of Cities of Italy: 16.2%
4. Country with the maximum number of cities: USA
```

In [ ]:
```python

```